Auditing Reference Distributions for Observation-Replacement Return Spans

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Code and data: https://github.com/Aaryansi/nano-l1

Abstract

An observation-replacement return span measures how a specified input intervention changes a policy's return. Interpreting that span requires choices about the reference population, replacement distribution and evaluation process. We audit these choices in a binary-contract simulator, controlled synthetic populations and six control-task configurations. We retain training-seed variation, raw endpoint returns, failed competence gates and unresolved references. In the historical market example, five policies have mean return -0.770 dollars per episode with a crossed seed/day interval $[-1.509, 0.011]$. Reference comparisons remain descriptive because market policies are not exchangeable with the synthetic bank. Across 20 independent synthetic reference banks, the zero-strength flag rate is 2.8% for each rule, but its conservative bank-level upper bound is 24.5%. This does not certify five-percent error control. The zero-strength population still has positive full-input replacement spans, illustrating that return sensitivity can arise without settlement-predictive inputs. An exact finite-pool example shows why the legacy reference-centered rank/normal rule lacks a general exchangeability guarantee; standard symmetric ranks address that separate statistical requirement. The audit supports careful interpretation of particular interventions, without inferring feature-level fidelity, absent information or explanation meaning from a single return-span comparison.

1 Question and scope

An explanation can describe a policy's decisions, its predictions, or the consequences of its actions. These are different targets, including when the policy performs poorly. SVERL makes this distinction explicit and assigns feature contributions to specified characteristic functions (Beechey et al., 2023, 2025). A faithful explanation of a costly decision can be useful for debugging it. Low return alone does not make the explanation empty.

Our question concerns a narrower diagnostic: how should an observed observation-replacement return span be compared with spans from a bank of reference policies? We examine the implemented comparison, the financial state retained by an intervention, and the evidence needed to interpret reference widths and classifications. The contributions are a reproducible computational audit, explicit examples of interpretation failures, and empirical comparisons under declared training budgets. We do not propose a universal null for reinforcement learning or a new general rank-test theorem.

The study is retrospective. Its configuration was fixed for the selected rerun after exploratory development on this historical data. Repeating that window does not create a prospective holdout. All numerical tables are generated from the selected primary release and two separately identified supplemental studies; the primary training source is `dcc4e3f31c69`. Reduced-budget execution runs cannot supply these tables.

2 What the endpoint statistic identifies

Let π be a fixed policy, M an environment, D an evaluation population, and I a specified replacement intervention. For a feature set S, let $v_{\pi,M,D,I}(S)$ denote expected episode return when the intervention retains the coordinates in S. The endpoint span is

$$\Delta(\pi, M, D, I) = v(N) - v(\emptyset). \tag{1}$$

We estimate both endpoints by replaying the same held-out episodes. The full endpoint supplies original policy inputs; the empty endpoint applies the declared replacement at every decision. The intervention also changes the policy's actions and subsequent inventory. Therefore this is a property of the policy, environment, intervention and evaluation distribution together.

For an efficient Shapley decomposition, $\sum_i \phi_i = v(N) - v(\emptyset)$ (Lundberg and Lee, 2017). Reading the endpoints avoids estimating individual coordinates merely to recover their sum. It does not validate that decomposition: allocations with the same sum receive the same diagnostic regardless of how they assign feature credit. Estimation accuracy for a ranking is another question; our implementation's gap diagnostic does not

implement the formal ranking procedures of Goldwasser and Hooker (2024).

The intervention is also distinct from SVERL's conditional policy-occupancy construction. The extended framework defines outcome explanations through partial-information behaviour at the state being explained, with the original policy elsewhere (Beechey et al., 2025). Our empirical flat-inventory background and repeated replacement at all steps differ in distribution and locality. The experiments do not test or refute that theoretical construction. FastSVERL provides practical approximations for these distinct behaviour, outcome and prediction targets, including sequential and off-policy estimation (Beechey and Şimşek, 2025). Our endpoint audit does not benchmark that method. Background selection and the relationship between SHAP feature importance and agent performance have also been studied empirically, including exceptions in fragile tasks (Engelhardt et al., 2024). We claim neither a new discovery of background sensitivity nor a general equivalence between importance and return.

Parameter sensitivity and probability attribution address separate targets; completeness does not make a probability difference a return intervention (Adebayo et al., 2018; Sundararajan et al., 2017).

3 Reference diagnostics and rank validity

The legacy rule compares an observed span T with B reference spans $T_1, \dots, T_B$. It combines a reference-centered empirical tail score

$$p_{\text{legacy}} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\{|T_b - \bar{T}_{\text{ref}}| \geq |T - \bar{T}_{\text{ref}}|\}}{B+1} \tag{2}$$

with a two-sided normal approximation to $z = (T - \bar{T}_{\text{ref}})/s_{\text{ref}}$. Both criteria must meet the declared level $\alpha = 0.05$. We retain this rule as a diagnostic of the original workflow, not as an assumption-free test. Insufficient rank resolution or $s_{\text{ref}} \leq 10^{-12}$ makes its comparison unresolved. This convention does not mean that every test against a point-mass reference is invalid.

Proposition 1. *The legacy combined rule does not have a general finite-sample exchangeability guarantee.*

Proof. Uniformly designate one observation in a fixed pool containing $-1, +1$ and 18 zeros; the remaining 19 entries are references. The resulting observation and references are jointly exchangeable. Either extreme receives legacy tail score $1/20 = 0.05$ and $|z| = 20/\sqrt{19} > 4.58$, so both meet the normal criterion. The rejection probability is $2/20 = 0.10$. This is a finite-pool example, not an error estimate for the independent synthetic benchmark. □

For comparison, let $T_0 = T$ and compute the pooled center $\bar{T}_{\text{pool}} = (B+1)^{-1} \sum_{b=0}^{B} T_b$. The symmetric two-sided rank uses scores $A_b = |T_b - \bar{T}_{\text{pool}}|$ and

$$p_{\text{sym}} = \frac{\sum_{b=0}^{B} \mathbf{1}\{A_b \geq A_0\}}{B+1}. \tag{3}$$

We also report the upper-tail raw-span rank. Both include the observation and count ties conservatively. Under joint exchangeability and permutation-equivariant scoring, their ranks are super-uniform by the established finite-rank argument (Kuchibhotla, 2020; Hemerik and Goeman, 2018). Conditional on an unordered score multiset, at most an α fraction of indices can have an inclusive upper-tail rank at most α; exchangeability makes the observed index uniform. The symmetric rank assigns each extreme in the example $2/20 = 0.10$.

That guarantee is marginal over the bank and the observed draw. It does not bound error conditional on each realized reference bank. It also does not supply exchangeability between a historical market agent and policies trained on synthetic populations or corrupted inputs. Connections between attribution and conditional-independence testing require explicit modeling assumptions, as in SHAP-XRT (Teneggi et al., 2023); the return setting does not inherit those guarantees automatically.

4 Implemented experiments

4.1 Market accounting and evaluation

The pinned corpus contains 6,425 binary-contract episodes with 14 decision rows each. Chronological training, validation and test counts are 3,854, 1,283 and 1,284, with four boundary episodes purged. The 18 inputs comprise 14 market/spot coordinates normalized using training statistics and four active-inventory coordinates. Actions target $-100, 0, +100$ signed yes-equivalent contracts. A reversal can trade 200.

The full-input replacement background contains 256 rows sampled from all evaluation steps while inventory is held flat. Each intervened episode draws a fresh whole background row at every decision. This replaces the clock as well as market and inventory coordinates; it is an empirical intervention, without a guarantee of compatibility with the recipient's state.

The assumed execution model buys at the supplied ask and sells at the bid. It charges the per-order fee $\text{ceil}_{\text{cents}}[0.07|C|p(1-p)]$, with zero impact by default, full fills and no funding or queue simulation. Candle-derived quotes are coarse proxies; these assumptions do not establish exact historical fees or live executable performance. The default terminal mark is the recorded binary settlement, with no additional settlement fee in the model. Equity increments telescope to final profit and loss at $\gamma = 1$.

Five market seeds train for 100 PPO updates under each cost objective. Their greedy policies are evaluated on every held-out episode. The critic predicts stochastic policy return in dollars, not settlement probability; its auxiliary calibration uses eight policy rollouts per episode. Both rollout implementations agree on every acted-on observation and greedy action for all ten checkpoints, with maximum accounting discrepancy $\leq 3 \times 10^{-14}$. Primary uncertainty crosses training-seed resampling with shared UTC-day blocks. There are only 14 test-day blocks, limiting inference about temporal variation.

4.2 Synthetic populations and reference banks

The balanced synthetic population has independent binary settlements, constant price 0.5, spread 0.02, and noisy features independent of settlement. Flat earns zero. The planted population adds a noisy settlement signal, repeated across the 14 steps. Its first-observation sign-and-hold strategy is a reference policy, not a proven optimum when aggregation, delayed entry and abstention are possible.

The primary bank has 24 independently seeded policies, each trained for 40 updates on 1,200 synthetic episodes and evaluated on an independent draw of 1,200 episodes with its own training normalizer. A 99-draw sensitivity retains the same procedure. The conditional calibration uses 30 independent observed trials at each strength $0, 0.2, 1$, reusing the primary bank. Each trial retains both endpoint returns. We report pointwise Wilson intervals conditional on that bank, alongside unresolved counts and both standard rank comparators. These rates include the specified learning procedure and budget. They do not isolate test sensitivity from learning success or quantify bank-to-bank error.

A separate study repeats the experiment across 20 independent banks, each with 24 references and 30 observed trials at each of the three strengths. All policies train for 40 updates on fresh 1,200-episode corpora, use their own training normalizer and evaluate on a fresh 1,200-episode corpus. Every policy has distinct data, training and masking seeds. Unlike the original conditional study, masking seeds are separate from policy seeds. This is a declared common procedure for all three rules, not a comparison isolating bank reuse alone.

The independent units are $K = 20$ banks. For bank flag fractions $X_k \in [0, 1]$, we estimate the marginal flag probability by their equally weighted mean. We invert the bounded-variable Chernoff bound (Hoeffding, 1963) with threshold $\log(2/0.05)/K$ for Bernoulli KL divergence. This yields conservative pointwise 95% intervals over banks, including at zero or unit observed rates. It does not treat the 600 trials as independent Bernoulli observations. Paired method differences use the same bound after mapping $[-1, 1]$ to $[0, 1]$. Intervals are not simultaneous over rules, strengths or contrasts. Unresolved comparisons remain in the operational flag-rate denominator and are reported separately.

4.3 Control tasks and competence gates

We evaluate CartPole, Acrobot, MountainCar, Pendulum with nine discrete torques, deterministic 4×4 FrozenLake, and CliffWalking capped at 200 steps. The control trainer is Stable-Baselines3 PPO (Raffin et al., 2021), with separate two-layer 64-unit actor/value trunks and four environment copies. Observed agents request 400,000 interactions; each blinded reference requests 30,000. Complete rollout batches can exceed those requests. Each construction has 24 reference draws, and endpoint evaluation uses 20 episodes. Five independently trained observed seeds are separate from correlated training checkpoints.

The parameter reference uses fresh orthogonally initialized actors, with no learning. The environment reference trains agents with independent coordinate-wise Gaussian observations whose moments are estimated from a random-policy rollout, leaving transitions and rewards intact. Both are evaluated on the original task with full inputs and with repeated background replacement; the background contains 512 random-policy observation rows. Gaussian blinding can violate within-state constraints. A separate whole-row resampling sensitivity preserves donor row support, without guaranteeing consistency with the recipient trajectory. Initialization and blinded learning are different reference procedures, without an established exchangeability relationship to competent observed agents.

Declared competence thresholds are respectively 475, -100, -110, -300, 0.8 and -20, measured on 25 episodes, together with an absolute measured endpoint span above 10^{-6}. A failed gate does not establish low detection power. Neither reference-width ratios nor standardized distances are power multipliers. All failed and unresolved cases remain in the results.

5 Empirical results

5.1 Historical market and primary references

The market policies have mean return -0.770 dollars per episode (Table 1). The primary centered-bootstrap probability is 0.0467. This approximate probability and the percentile interval are different summaries and need not be dual near the decision level. We report both, without selecting one to support a stronger claim. Neither a negative mean nor a comparison with flat proves absence of predictive information or makes a policy's explanation meaningless.

Table 1: Held-out market return in dollars per episode. Seed spread is the sample SD across seed means, not episode SD or a standard error. Cost and zero-cost agents were trained separately. The primary crossed seed/day interval is [−1.509, 0.011]. Non-rejection would not establish equivalence.

Training objective	Seeds	Mean	Seed SD
Fees and spread	5	−0.770	0.354
Zero cost	5	0.056	0.984

Table 2: Primary synthetic-bank diagnostics. "Outside" denotes the legacy two-part classification, not a finding of explanation fidelity. Market comparisons are descriptive. Undefined distances are shown explicitly.

Case	Span	Ref. SD	z	Decision
Planted signal	75.627	3.857	17.61	outside
Zero strength	12.436	3.857	1.23	within
Market seed 0	8.848	3.857	0.30	within

Table 2 compares the fixed primary cases against the declared bank. All five market-seed comparisons and the larger-bank sensitivity are retained in the release. Across 5 market seeds, the primary bank gives 0 outside, 5 within and 0 unresolved classifications. Their spans range from 7.409 to 11.846. The larger bank gives 0 outside, 5 within and 0 unresolved classifications across the same 5 market seeds. Changing a bank can change its center, width and classification while also changing training or evaluation distributions. Those differences do not identify a unique information-removal mechanism.

5.2 Conditional and marginal calibration

At zero strength, the legacy rule flags 0/30 resolved trials (conditional interval [0.000, 0.114]), with 0 unresolved trials. The upper-tail and symmetric ranks flag 0/30 and 0/30, respectively. Their conditional intervals are [0.000, 0.114] and [0.000, 0.114]. Figure 1 plots these rates and their intervals. With 30 trials, even zero flags leave a substantial upper uncertainty bound. One positive-control success is therefore not a power estimate. The signal strength sweep also changes what the learning procedure can acquire within its fixed budget.

Table 3 adds bank-to-bank variation to the audit. The zero-strength rate is 2.8%, but its upper bound is 24.5%. The observed rate below 5% does not certify error control at 5%. All three methods give identical trial outcomes across all 1,800 trials. This is empirical agreement, not equivalence. The pointwise paired-difference intervals extend as far as [−0.555, 0.555] in probability units. This empirical agreement does not contradict the finite-pool counterexample: that example rules out a

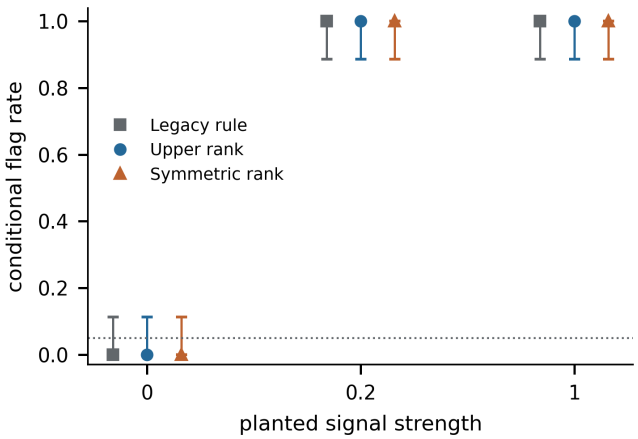


Figure 1: Conditional trial rates with pointwise intervals for one reused reference bank. A standard rank guarantee under exchangeability is marginal over banks; these intervals do not establish such a guarantee on the market.

general guarantee for the legacy rule, rather than predicting its error in every population. The bank study retains all 2,280 checkpoints and update logs; both endpoints were reproduced from every saved policy after extraction from the distributed archive.

5.3 Matched objective replication

The same market episodes and state transitions support two training objectives: trading profit in dollars, or the sum of signed settlement calls at each step, with no execution costs in the prediction score. Both objectives train from scratch at 40 and 100 updates. Each cell has five observed policies and 24 Gaussian-blinded references trained under that cell's objective and budget. The normalizer uses the original training split, and every policy is evaluated on the complete test partition with both full and replaced observations. This study uses a 192-row flat-inventory background, common masking draws across observed agents and reference-specific masking draws. Its replacement background and draws differ from the primary 256-row comparison, so those span differences cannot be attributed to the objective alone. Across the objective cells, 20/20 observed agents are outside their respective legacy reference, with 0 unresolved comparisons. Table 4 shows why that shared classification does not identify a shared mechanism: reference widths and observed spans change with the task and budget. The supplemental replication retains raw full and masked endpoints for 20 observed and 96 reference policies. Every original full-return array, observed/reference span, finite comparison and decision reproduces exactly (maximum episode-return difference 0.000000000). This reconstructs the new replication's masked endpoints; it does not recover missing original arrays or replay original objective checkpoints. No

Table 3: Independent-bank calibration: 20 banks with 30 trials per strength in each bank. Rates and pointwise bounded-bank intervals are rounded to three decimals. All declared trials remain in the denominator. Unres. counts unresolved comparisons. The signal alternatives measure learning and comparison together, not test power in isolation.

Signal	Rule	Flags	Rate	95% CI	Unres.
0.0	Legacy	17/600	0.028	[0.000, 0.245]	0
0.0	Upper rank	17/600	0.028	[0.000, 0.245]	0
0.0	Symmetric	17/600	0.028	[0.000, 0.245]	0
0.2	Legacy	575/600	0.958	[0.729, 1.000]	0
0.2	Upper rank	575/600	0.958	[0.729, 1.000]	0
0.2	Symmetric	575/600	0.958	[0.729, 1.000]	0
1.0	Legacy	600/600	1.000	[0.832, 1.000]	0
1.0	Upper rank	600/600	1.000	[0.832, 1.000]	0
1.0	Symmetric	600/600	1.000	[0.832, 1.000]	0

Table 4: Matched-budget objective cells. Return and span are means over five observed seeds. Reference SD describes 24 blinded policies. Prediction uses signed-call score units; trading uses dollars. Cross-objective magnitudes and standardized distances do not measure a causal objective effect.

Updates	Objective	Return	Span	Ref. SD
40	Prediction	7.319	7.301	2.451
40	Trading	−2.951	9.495	1.836
100	Prediction	7.386	7.384	2.373
100	Trading	−0.770	6.946	0.135

Table 5: Independent control seeds. Eligibility requires both the declared return gate and measured endpoint dependence. Classification counts retain all seeds and report unresolved references separately. Each task uses its native reward units. O/U denotes outside/unresolved counts out of five seeds.

Task	Return	Elig.	Env. O/U	Param. O/U
CartPole	500.0	5/5	5/0	1/0
Acrobot	−75.8	5/5	5/0	5/0
MountainCar	−185.4	0/5	0/5	0/5
Pendulum	−166.9	5/5	5/0	5/0
FrozenLake	1.0	5/5	5/0	5/0
CliffWalking	−200.0	0/5	0/5	0/0

objective-effect significance or explanation-fidelity claim follows from these comparisons.

5.4 Control-task competence and references

Of 30 independently trained control agents, 20 meet both eligibility gates. Of those, 20 have two resolved references; their classifications differ in 4 cases. No unresolved comparison is counted as reference agreement. Table 5 supports statements about these configurations and reference banks. It does not establish a universal ranking of reference constructions. A reference can collapse when its policies have identical returns, and the legacy normal-scale rule then cannot decide. That outcome differs from a resolved non-rejection and from a trained observed agent failing to learn.

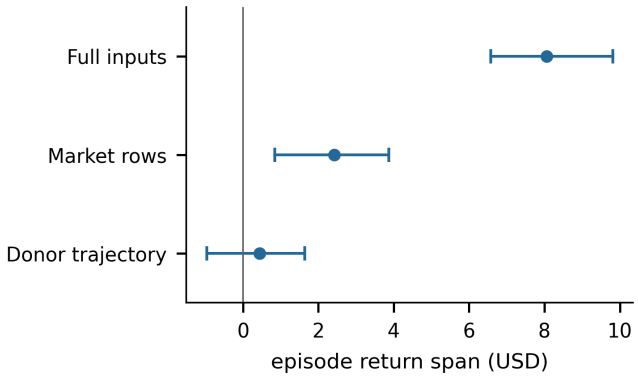


Figure 2: Market return spans under three specified interventions. Intervals cross training seeds and common UTC-day blocks, conditional on fixed masking draws. Differences do not isolate inventory as their sole cause.

5.5 State preservation and attribution suppression

The market state controls use the five saved cost-enabled checkpoints. Each synthetic case trains five new policies for 60 updates, using 1,200 episodes per training draw and 1,200 common independent evaluation episodes. Full replacement changes inventory inputs as well as market inputs. The market-row and donor-trajectory controls retain the real clock and all four inventory coordinates of the intervened rollout. A donor row preserves empirical row support, and a donor trajectory preserves its own market sequence. Neither guarantees joint consistency with the recipient's financial state or reward.

The full-input replacement span is 8.065, with interval [6.579, 9.816]; the donor-trajectory span retaining in-

ventory and clock is 0.443, with interval [−0.963, 1.634]. Figure 2 compares the three interventions. These runs fix masking draws within each analysis; their recorded seeds differ from the primary comparison's draws for some policies. The zero-strength population also has a full-input span of 9.859, with interval [4.075, 16.183]; its donor-trajectory span is −0.118, with interval [−0.795, 0.362]. Positive sensitivity in this generator need not indicate settlement-predictive information: replacement can change trading and its costs. These controls do not isolate one cause of the span differences.

Retaining inventory also retains some price information. If $z = q/100$ is scaled nonzero position, e its entry price and $u = q(P-e)/100$ scaled unrealized profit, then

$$P = e + u/z. \tag{4}$$

The identity holds up to feature precision. Thus a financially consistent control retaining these coordinates cannot also hide all current-price information when $z \neq 0$. Conditional information beyond retained inventory can still be a legitimate target; this identity is not an impossibility result for conditional explanations.

The steering experiment penalizes input gradients during training and measures the resulting behaviour-attribution share and held-out return separately. An auxiliary single-seed comparison examines greedy behaviour; it cannot establish behavioural equivalence across the full experiment. The penalty targets the expiry clock in the market and the planted signal in the synthetic case, with ten seeds and penalty 20 fixed for the reported comparison. Each steering policy trains for 60 updates. Synthetic training and independent evaluation each use 1,500 episodes; market evaluation uses the complete test partition. All penalties are retained and shown in Figure 3. For the market clock, the mean target share changes from 43.0% to 2.9%; the paired return change is −0.072 dollars, with interval [−1.072, 1.066]. For the synthetic signal, the mean target share changes from 44.7% to 2.4%; the paired return change is −58.098 dollars, with interval [−64.563, −52.458]. A non-significant return contrast is not equivalence or noninferiority, and changed task return alone cannot establish that the resulting policies are behaviourally identical.

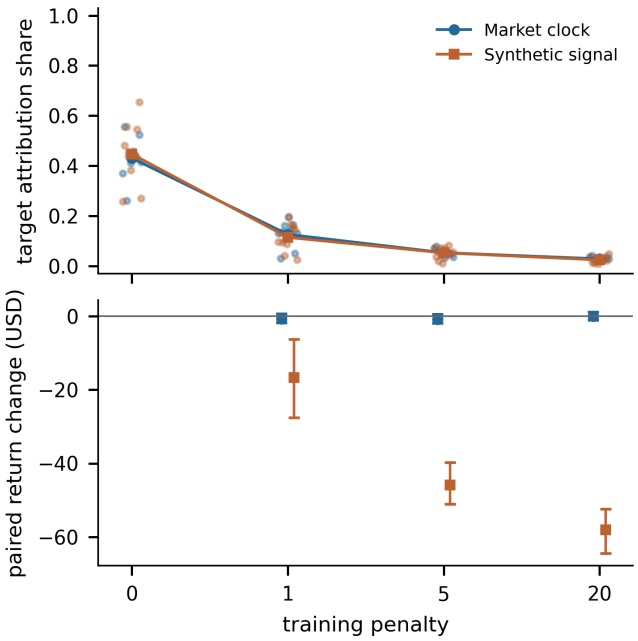


Figure 3: All declared penalties. Upper panel shows seed-specific target shares and their means; lower panel shows paired return changes with crossed seed/block intervals. The two populations target different features.

6 Limitations and next experiments

The financial model uses assumed costs and coarse quotes, the market period is fixed, and seed/day inference rests on few time blocks. Synthetic calibration includes a conditional study and an independent-bank study, but still has limited precision at rare error rates. Control competence is configuration-specific; failed tasks remain evidence about training rather than about detection sensitivity. No reference width alone identifies an explanatory mechanism.

Complete replay and hash checks verify computations, without establishing feature-level fidelity. Additional raw evidence and implementation limits remain documented in the full profile and release.

The next useful experiments are competence-controlled reference comparisons with matched learning budgets and conditional path randomization in a simulator with a known joint law. Such a benchmark should assess individual feature allocations, including deliberately incorrect allocations with the same endpoint sum. A larger independent-bank study should choose its sample size and precision target before further runs; extra trials within existing banks do not create additional independent banks. A separately frozen temporal market holdout would test external robustness. These additions would strengthen calibration and generalization; they are not prerequisites for honestly reporting the present descriptive audit.

Reproducibility. The repository pins the corpus and splits, retains raw returns and model hashes, records numerical runtime and package versions, and verifies source against the training commit. The full runner and release verifier are separate from execution-only smoke checks. Manuscript tables, figures and summaries share one derived statistics artifact; the three document profiles share this source.

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